# MycoEdge

ONE LIFETIME RAZOR, ENDLESS GROOMING.




**SUSTAINABLE DESIGN ENGINEERING**

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# PROBLEM & RESEARCH INSIGHTS

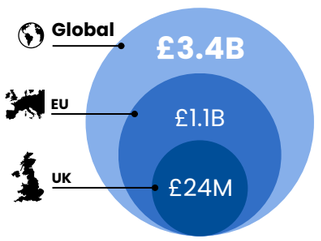
## Context

This report aims to summarise the key problems and insights regarding the razors' FMCG market, and propose a **Design Engineering solution that reduces the wet shaving industry's environmental impact**.

### Economic [1]

In a £3.4B market, expected to grow at a rate of 3.6% reaching **£4.2B by 2030**, the environmental impact can potentially be severe.

Market Size




Market Growth & Users

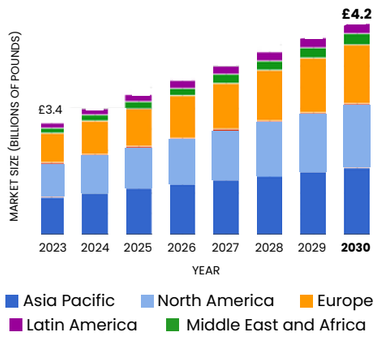
**+3.6%** CAGR

**33.7M**

Potential Users - UK

Over 33M customers drive this market through **increasing personal grooming trends**. Manufacturers play a significant role as well, being **pressured to adopt greener approaches through organic components** amid raising environmental concerns & regulations.

### Social [2]

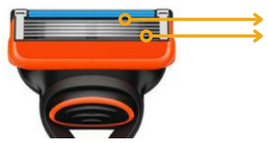



**Coorporate culture in Europe & North America** places raising emphasis on **men's personal grooming**, including changing beauty standards and skincare trends.

**73%** Of **global consumers are willing to change their consumption habits** to reduce their environmental impact per Nielsen

## Environmental Issues

### Inefficient Design for Disassembly



Only 2/7 Components Fail:

- Stainless Steel Blades
- Lubrastrip

The remaining **5 components are fully functional** when disposing of the cartrigde. It is a waste of resources because **no cartridge design allows for maintainance** or partial component replacement.

### Recycling Challenges [3]



- **< 40x40 mm components are not sorted**, thus not recycled and disposed of to landfills.
- **Hazardous diseases or pathogens in blades** require specialised recycling treatment.

### Excessive Water Consumption [4]




Water used in **shower** shaving session

Water used in **sink** shaving session

Water usage in the wet shaving industry is a **significant, often overlooked issue**. A user can use up to 85L of water per shave, posing a threat to natural resources.

### High Carbon Footprint [5]

A single **razor produces 4.68 kg CO2 / year**. When multiplied by the number of users, and divided by the total amount of UK CO2 emissions in 2022 (**331.5M Tonnes**), we get a total emission contribution of:

Total CO2 kg Emissions Per Year

**157.4 T**

Contribution to UK CO2 Emissions Per Year

**0.47%**

## Users

### Needs

Simple, affordable, efficient personal grooming routine with wet shaving

Convenient purchase options for recurrent orders

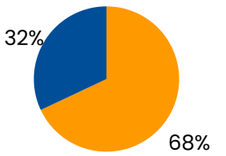
### Pain Points

Dilema between finance and sustainability: eco-friendly products are more expensive

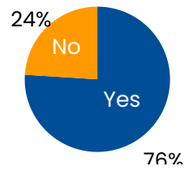
Greenwashing: hard to find truly sustainable and skin-kind products

A survey was conducted with 50 individuals. Results show that **quality & price are two aspects to overcome** in the green era:

Main Incentive...
Price or Sustainability?




Eco-friendly materials as lower quality?




## Insights & Opportunities

**A. Affordability as the Gateway for Environmental Engagement:** With high inflation (7.6%), consumers may opt for cheaper, less eco-friendly shaving products.

B. **DfD & Recycling Issues Hinder Materal Circularity.** No maintenance options for blades, the cartridge's small size and contagious diseases in blades pose a threat for a close loop.

C. **Wet shaving sessions**, particularly during showers, can consume up to 85 liters of water, threatening natural resources.

**Final Brief:** HMW redesign affordable and maintainable razors to enable material circularity while aligning with stakeholders' eco-values and incentives?



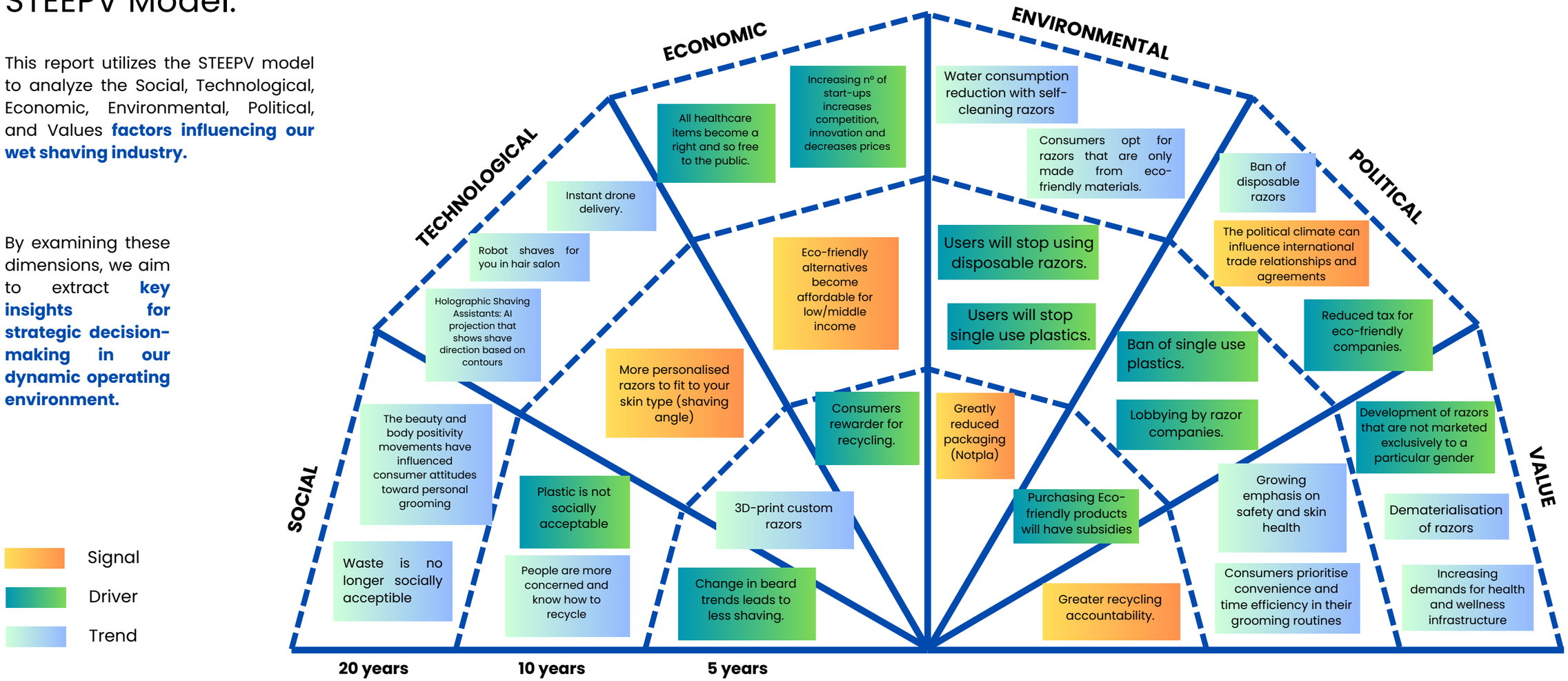
**References:** [1] https://www.investopedia.com/terms/f/fastmoving-consumer-goods-fmcg.asp [2] https://www.nielsen.com/es/
[3] Dr Marco Aurisicchio. Sustainable Design Engineering Week 3 Lecture Notes [4]: https://www.waterwise.org.uk/save-water
[5]: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1147372/2022_Provisional_emissions_statistics_report.pdf

# FUTURE SCENARIO & TIME HORIZON

## STEEPV Model:

This report utilizes the STEEPV model to analyze the Social, Technological, Economic, Environmental, Political, and Values **factors influencing our wet shaving industry.**

By examining these dimensions, we aim to extract **key insights for strategic decision-making in our dynamic operating environment.**




## What May Happen in...

### 5 Years

Expect blade technology advancements and **smart razors with IoT(Internet of Thing) for personalized shaving.**

Plastic waste concerns may boost the use of recycled materials, while **convenience and affordability will likely drive market trends.**

### 10 Years

Anticipate **biodegradable razors and nanotech-driven precision blades**.

The market may shift towards sustainability, favoring eco-friendly, long-lasting options. Inclusive marketing strategies could reflect **evolving societal norms.**

### 20 Years

**AI and robotics** integration could redefine shaving.

**Smart razors adapting to individual features may emerge**. Traditional disposable razors might decline due to heightened concerns about plastic waste.

### Key Points

- Smart razors with IoT capabilities
- Integration of AI improves shaving UX and personalisation
- More biodegradable materials options
- Disposable razors disapear
- Change in grooming trends

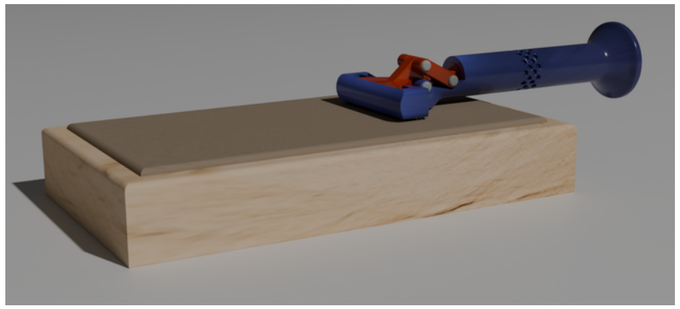
# PRODUCT & PACKAGING SOLUTION DESIGN

## Our Solution:

**The MycoEdge razor is designed to last a lifetime.** It is **monomaterial**, **made out of 304 Stainless Steel**, which is a corrosion resistant, high strength, responsability sourced material. The user can **sharpen the blades multiple times with a built-in whestone**. Each time a blade gets sharpened, the user twists the bottom of the handle. This **L-lever pushes the blade forward to account for the reduction in length**. Furthermore, we implemented a **vertical standing feature as it enhances drying** and storage, expanding the lifespan even more (per Gillette).

## Sharpening the Blades:

The aim of our product is to **allow the user to sharpen their own razor blades**. To do this, the packaging has a **built-in whetstone** on the back which the razors are passed across. This results in them being worn down over time which is why we have a mechanism that pushes them forwards slowly.



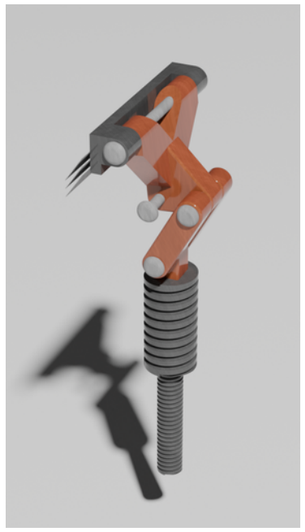



## The Mechanism:

The **mechanism works to push the blades forwards after sharpened** as the bottom of the handle is **twisted**.

This turns a screw against the handle which pushes it upwards onto an **L-lever** which **translates the movement into the razor blades** themseves.

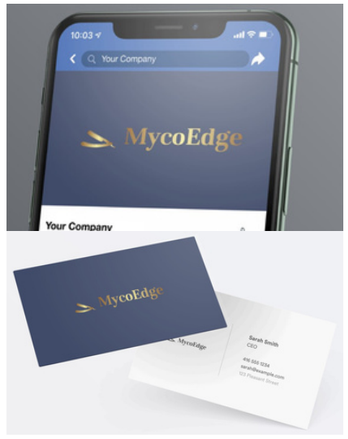
These then move along channels at a **35°** angle, the ideal shaving angle.



## The Brand:

At MycoEdge, our **values** are deeply rooted in **sustainability, durability, and the timeless art of grooming.**

MycoEdge is not just a razor; it's a symbol of how **disruptive innovation and high-quality can drive a greener future** in the fast-moving consumer goods landscape while ensuring a **viable business model**.




## Mycelium Packaging:

Mycelium packaging utilizes the root structure of fungi to bind agricultural waste, like corn husks, creating a **naturally sourced alternative**. Its unique growth process forms **resilient**, **customizable** packaging that **biodegrades** and is **home compostable.**




As a **natural material from a renewable source**, it is **easy to grow, needing very little water, light or space** to produce at an industrial scale. Moreover, its **lightweight nature reduces transportation emissions**, contributing to a more eco-efficient supply chain.

Aside from sustainability, Mycelium has been proved to be "**high-performing, cost-competitive, and insulating protective packaging solution**." [22]



**References:** [22]: https://mushroompackaging.com/ https://www.sourcegreen.co/food-packaging/mycelium-packaging-styrofoam-alternative/

# SYSTEM SOLUTION DESIGN

MycoEdge offers a system that **ensures responsible mining and sourcing**, **sales and profit for the business**, **water usage reduction**, **material circularity & recycling** as well as a **significant extension in product life**. Thus, provides a **win-win scenario for all stakeholders and the environment**.




## 1. Materials Supply

[30] outokumpu

The company provides stainless steel in sheets and bars.

ECOVATIVE DESIGN

A biotech company that designs packages using mycelium technology.

[32] SUEHIRO

The company manufactures whetstones by sedimentary rocks, like siliceous mudstones and volcanic ash deposits

## 2. Manufacturing & Processing

[23] kai group

New stainless steel blades are manufactured.

[24] JC METALWORKS

The stainless steel bar is manufactured into the razor handle.

Assembly

Stainless steel sheets are transported by eco-trucks

Distributed to the warehouse -> Delivered to the store

## 3. Online and Offline Sales

[25] Boots — Offline

[26] a — Online

Package

Amazon trucks

Blade replacement

## 4. Product Service System & Consumption

Refill

**Consumer**

Chronologically

1. Unpack
2. Shave
3. Rinse
4. Sharpening

Biodegradable packaging

The blades are too short for sharpening

Decade lifetime. When obsolescence, it is recyclable

Natural stones: no need to process before obsolescence

Water used when shaving and rinsing

[29] Thames Water

The public utility that provides wastewater service.

[28] FCC Environment

The company collects the general waste and processes before landfill.

## 5. Dealing with Waste

[27] SIMS METAL

The company recycles the stainless steel blades.

**Closing the Loop**

The recycled stainless steel is purchased and delivered to the company as the source of material.



**References:** [23]: https://www.kai-group.com/global/en/kai-factory/process/razor/ [24]: https://www.jcmetalworks.co.uk/metal-manufacturing-services/ [25]: https://www.boots.com/ [26]:https://www.amazon.co.uk/ [27]: https://www.simsmm.com/about-us/ [28]: https://www.fccenvironment.co.uk/ [29]: https://www.thameswater.co.uk [30]: /https://www.outokumpu.com/en [31]: https://mushroompackaging.com/ [32]: https://www.suehiro-toishi.com/en/

# INFRASTRUCTURE

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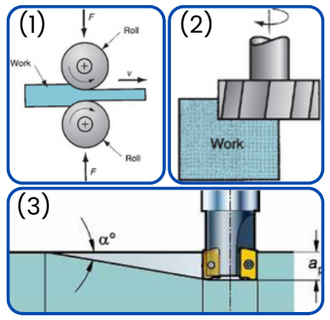
## Manufacturing

### Materials [33][34]

This product is designed to be mono material to **ease the disassembly in recycling process**.

### Stainless steel

- Rolling (1)
- Stamping
- Heat treatment
- Edge milling (2)
- Honing (3)
- Coating

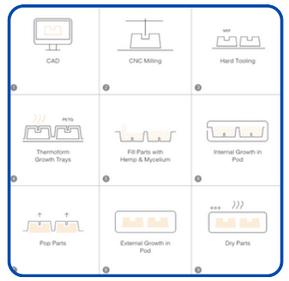



### Packages [31]

The package is made by mycelium which is **completely biodegradable and compostable.**

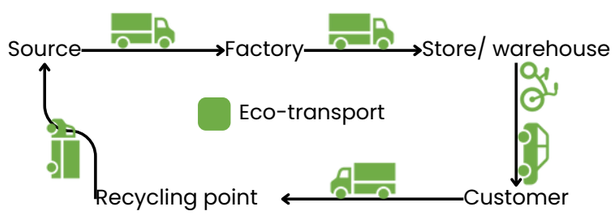
### Mycelium and hemp hurd

- CAD
- CNC Milling
- Hard tooling
- Thermoform growth trays
- Growth
- Dry




## Transport

- Making the product **size compact** (the razor and the six blade refills are embedded in the package.) which **saves the space** for lorry delivery.
- **Streamlining distribution** to **reduce the emission** related with transportations.
- Example:



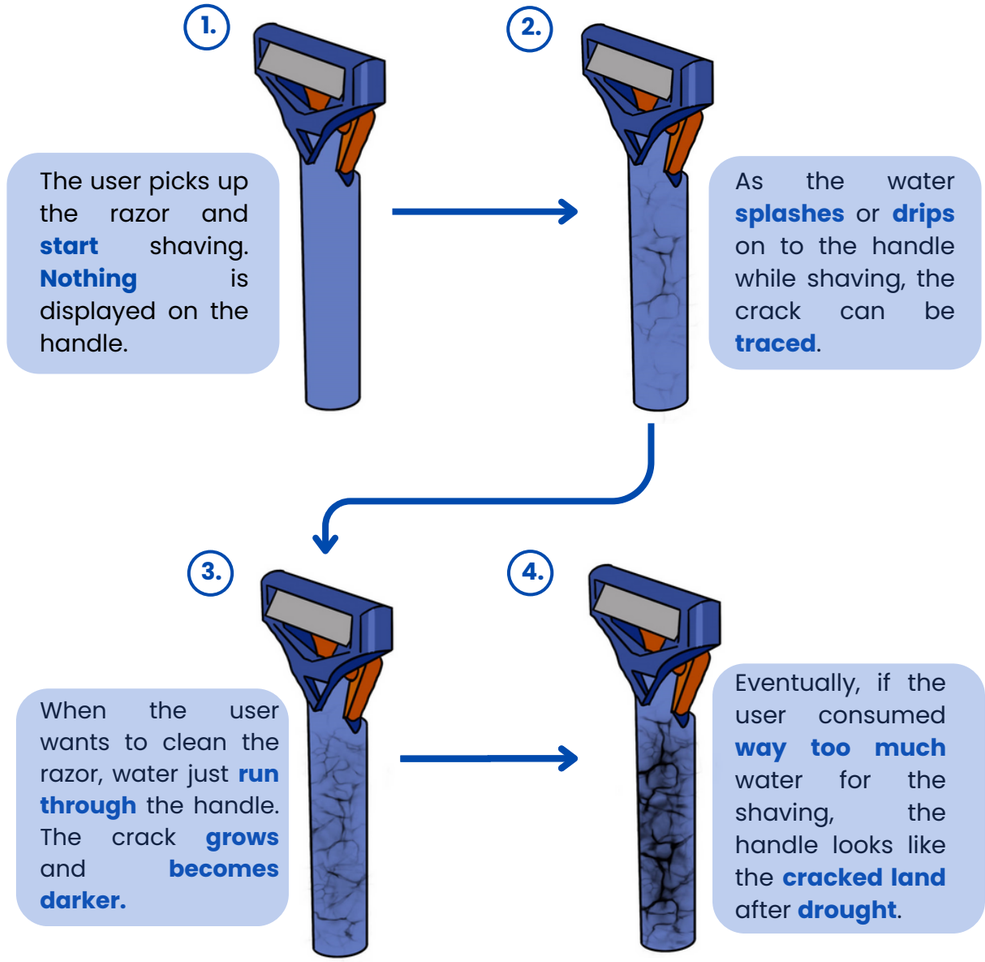

## Operation

### Shaving Issue [Research Insight A]

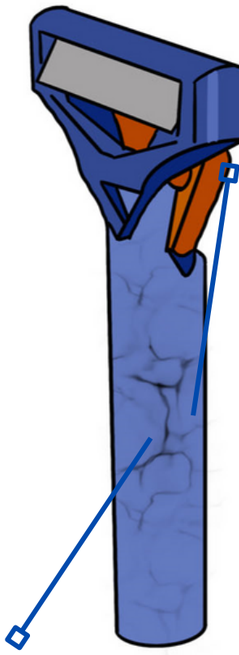
Wet shaving consumes **a huge amount of water** which threatens the natural resources.

### Extra Feature: Aqua Crack Clock

This extra feature is applied to the handle of the razor to **remind** people **how much water they have consumed** for this shave visually.




### Hydrochromic Ink



The handle is manufactured with **hydrochromic** ink in the shape of cracks.

The ink **changes colour** as water is **in contact with it**. [35]

The hydrochromic inks is **environmentally friendly** and **hazardless**, as it can be used for baby's dipper



**References:** [33]: https://barbersets.com/blogs/blogs/how-are-razor-blades-sharpened-and-manufactured [34]: Dr Billy Wu Manufacture slides [35]: https://www.matsui-color.com/ourproducts/specialty-products/aqua-indicator

# INFRASTRUCTURE

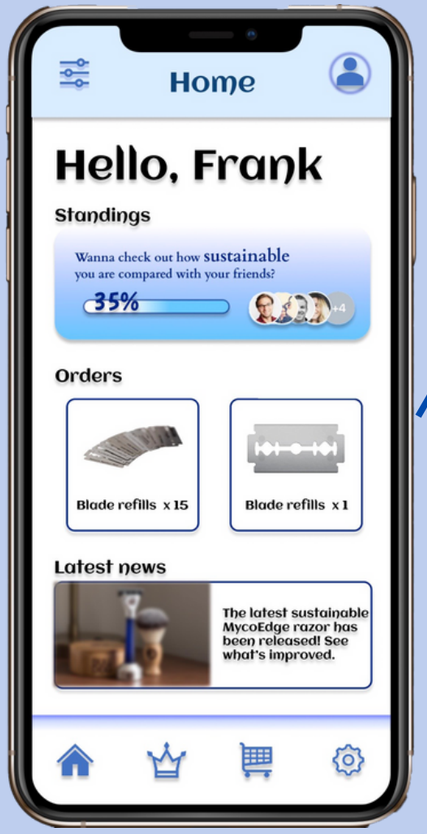
TIANLE YAN

## MycoEdge app

It is a **versatile** app with functions:

- Shop for shaving products
- Recycle blades for discounts, and grow your virtual tree
- Compare tree growth with friends and track your impact,
- Join a global community promoting sustainable shaving practices.

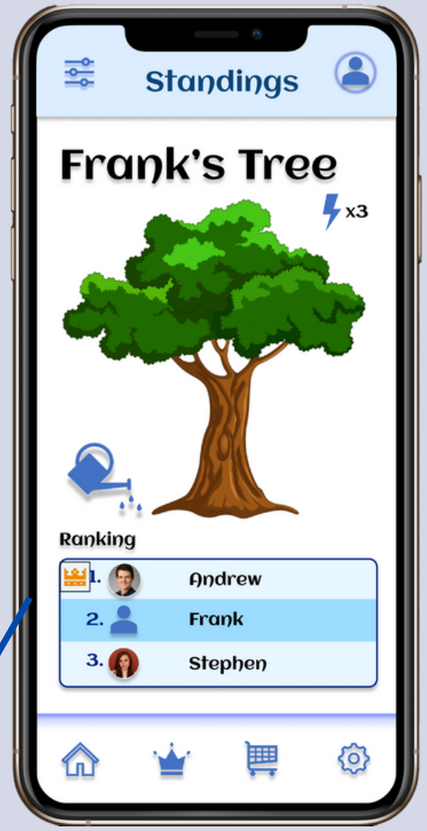
### Home page




The home page of the MycoEdge application consists of some **brief information** about the **standings**, the **orders** and **latest news** which user may be interested at.

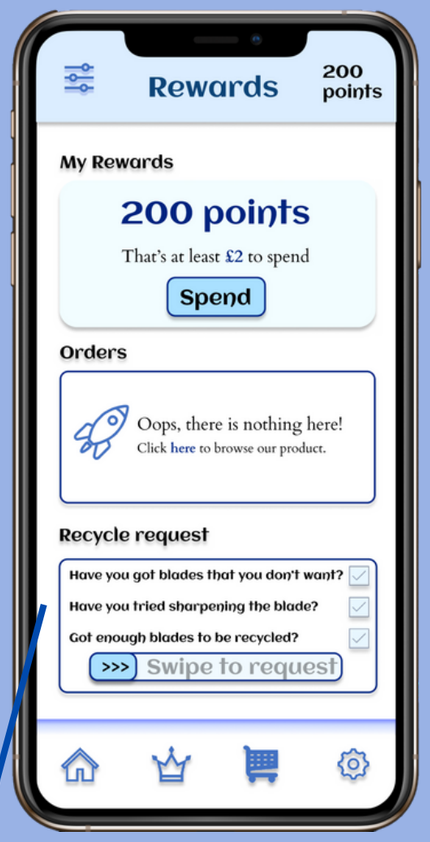
This section indicates how **sustainable** you are compared with your friends.

### Interacting page




This page shows a **virtual tree** that user can interact with (watering the tree). It takes **energy** (the key on the top-right corner) each time. The energy can be obtained by **recycling** the blades with MycoEdge's partners.

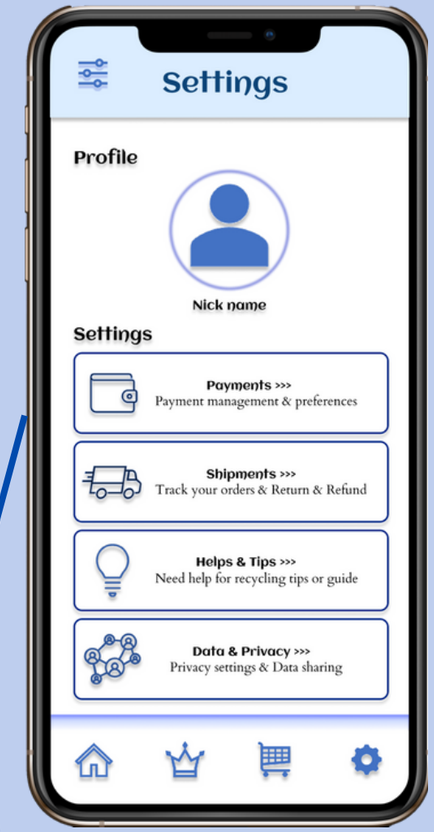
### Rewards page




The rewards page illustrates the **points** you had from recycling the blades with MycoEdge's partners, and the **equivalent values of voucher** you can use to order blade refills next time.

The bottom section is for user to request a **pick-up service to recycle** the blades from user. The tick lists are there to **guide** user before they **obsolete** the blades.

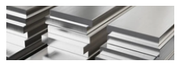
### Profile page




The setting page is where user can put their **personal information** down and change some **preference settings** during the use of the MycoEdge app.

# Resource Data: Why Stainless Steel?

**Our prime objective was designing a monomaterial razor**. Current products have a combination of elastomers, polymers and metals that **hinder recyclability**. We compared CO2 emissions of:

| | CO2/kg Material | Razor Weight | Lifespan & n° Razors Used | CO2/User | CO2 in UK |
|---|---|---|---|---|---|
| **Stainless Steel**  | **2.48 kg CO2** [36]<br>**1 kg Stainless Steel** | 80 grams<br>0.2 kg CO2 | 10 years - 6 razors | 1.24 kg | **41.5M kg** |
| **PLA**  | 0.5 kg CO2 [37]<br>1 kg PLA | 40 grams<br>0.02kg CO2 | 1 month - 720 razors | 14.4 kg | 480.9M kg |
| **Aluminium**  | 4.0 kg CO2 [38]<br>1 kg Aluminium | Discarded: Corrodes if water PH is between 4.5-8.5 (tap water is pH 6.5-9.5) | | | |
| Statistics & Assumptions | Includes all process: mining, sourcing, transportation and manufacturing | Based on material's density and market products' volume and weight | Based on market products' longevity claims and reviews. Assume shaving lifespan ranges from **20 to 80 years old** | Assumes that user's shaving frequency remains constant | From research slide, **33.4M potential users** in the UK |

Altough PLA seemed more eco-friendly by having a lower CO2/kg of material, the **long-lasting nature of stainless steel** made it the best in the sustainability, price, user experience & brand trust aspect.

**Advantages:**

- **Sustainable: - 91.7% CO2** with Stainless Steel
- **Cheaper: - 93.3% Materials Cost** (£15/kg PLA vs. £1/kg Stainless Steel)
- **Improved UX:** Stronger grip and higher perceived quality
- **Increased Brand Trust:** more customer faith with robust, long-lasting product

# Impact Data: Embodied Energy Eco-Audit

Our product's **lifetime environmental impact** was **analysed through CES Edupack** and compared to the leading product in the market: Gillette Fusion 5, which has multiple eco-friendly claims.

## Results & Solution Advantages

**-65% Total Embodied Energy**

### -65.4% in Materials

A long-lasting stainless steel razor is more eco-friendly that a combination of elastomer/polymers

### -23% in Transport

Gillette ships its products from Berlin, while our Outokumpu factory is in Sheffield (800km less).

### -91% in EoL Potential

Monomaterial feature allows for full recycling, whereas small multimaterial components do not

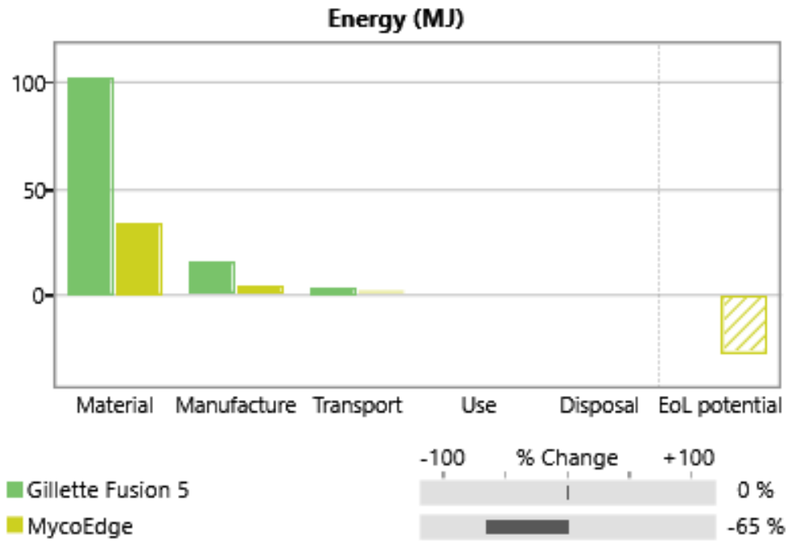
## Analysis

Over a lifespan of 60 years, a Fusion5 user uses 60 handles and 720 cartridges. A MycoEdge uses only 6 handles and re-sharpens its blades. Thus, **reduction in material sourcing, manufacturing, transportation and waste.**

Recycling is a major aspect. Fusion5 integrates elastomers within its metalic handle, and polymer within its blades, which impeeds recycling. A fully **stainless steel razor can be melted and reused** for the same application.

**Assumptions:**

- 30% scrap material used for both cases in stainless steel
- Elastomer & polymer components in Fusion5 cannot be recycled due to small size
- Terracycle is not used in Fusion5 due to lack of user incentives
- Transport for material sourcing was assumed negligible.






**References:** [36]: https://www.researchgate.net/figure/Climate-impact-in-kg-CO2-eq-per-kg-of-stainless-steel-tube-with-the-scrap-input-ratio_fig29_344364006
[37]: https://www.totalenergies-corbion.com/news/low-carbon-footprint-of-pla-confirmed-by-peer-reviewed-life-cycle-assessment/
[38]: https://www.hydro.com/en/about-hydro/stories-by-hydro/reduce-your-carbon-footprint-with-low-carbon-aluminium/#:~:text=Measured%20from%20start%20to%20finish,from%203.4%20to%203.9%20kgs.

JAVIER DE LA FUENTE GUILLEN

# Supply Chain Stakeholders

### Mycelium Packaging - Ecovative [39]




Ecovative is the **leader in the niche mycelium market**, backed up with **open source patents & strong partnerships** such as BESTSELLER.

Utilizing the natural growth properties of mycelium, the root structure of fungi, **Ecovative crafts biodegradable and compostable packaging**. It offers functional protection for products but also supports a closed loop.

### Stainless Steel - Outokumpu [40]




Outokumpu is the only industry leader in the UK that has been **accredited for sustainable sourcing and mining**, and invited to the **United Nations Climate Change Conference (COP28).**

Its reputation is attributed to its combination of **product quality, diverse offerings, competitive pricing** and commitment to sustainability in a **reliable supply chain.**

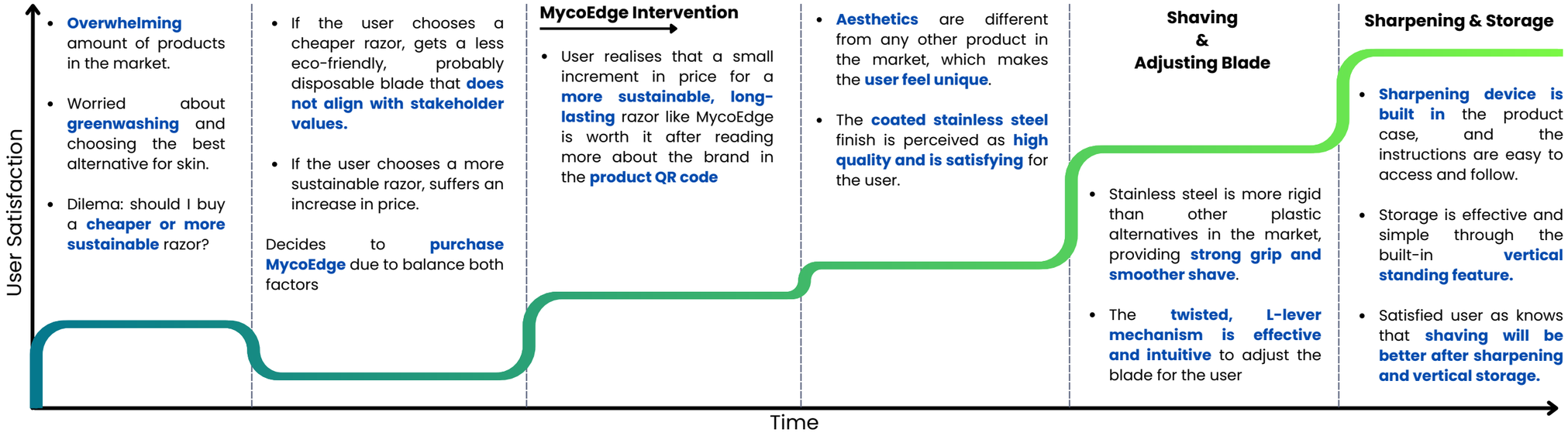
# User Experience Journey

After research, ideation and user testing, the following map was created to showcase the different product usage stages and how they evoke certain emotions or feelings within the actors:

MycoEdge

| | Pre-Aquisition | Aquisition | Post-Aquisition | Pre-Utilisation | Utilisation | Post-Utilisation |
|---|---|---|---|---|---|---|
| **Interaction & Behaviours** | • **Overwhelming** amount of products in the market.<br>• Worried about **greenwashing** and choosing the best alternative for skin.<br>• Dilema: should I buy a **cheaper or more sustainable** razor? | • If the user chooses a cheaper razor, gets a less eco-friendly, probably disposable blade that **does not align with stakeholder values.**<br>• If the user chooses a more sustainable razor, suffers an increase in price.<br>Decides to **purchase MycoEdge** due to balance both factors | **MycoEdge Intervention**<br>• User realises that a small increment in price for a **more sustainable, long-lasting** razor like MycoEdge is worth it after reading more about the brand in the **product QR code** | • **Aesthetics** are different from any other product in the market, which makes the **user feel unique**.<br>• The **coated stainless steel** finish is perceived as **high quality and is satisfying** for the user. | **Shaving & Adjusting Blade**<br>• Stainless steel is more rigid than other plastic alternatives in the market, providing **strong grip and smoother shave**.<br>• The **twisted, L-lever mechanism is effective and intuitive** to adjust the blade for the user | **Sharpening & Storage**<br>• **Sharpening device is built in** the product case, and the instructions are easy to access and follow.<br>• Storage is effective and simple through the built-in **vertical standing feature.**<br>• Satisfied user as knows that **shaving will be better after sharpening and vertical storage.** |
| **Motivations, Incentives & Investments** | User **invests time and mental energy** in the research and product selection. The driver is the personal grooming upgrade. | User is **motivated to fulfill their sustainable or price values** with the purchase. Can be incentivised by appealing packaging or special offers. | User invests more budget into our product in exchange for **more value and sustainable practice** in the long run. **Incentivised by one-time purchase.** | User starts to feel the **return on investment** by the perceived high-quality of the product and aesthetically pleasing finish. | **A more rigid handle incentivices** the user because it **produces less nicks and cuts**. User invests time and attention into the shaving process. | Sharpening the blade and **not having to buy new** ones motivates the user as it saves time. |

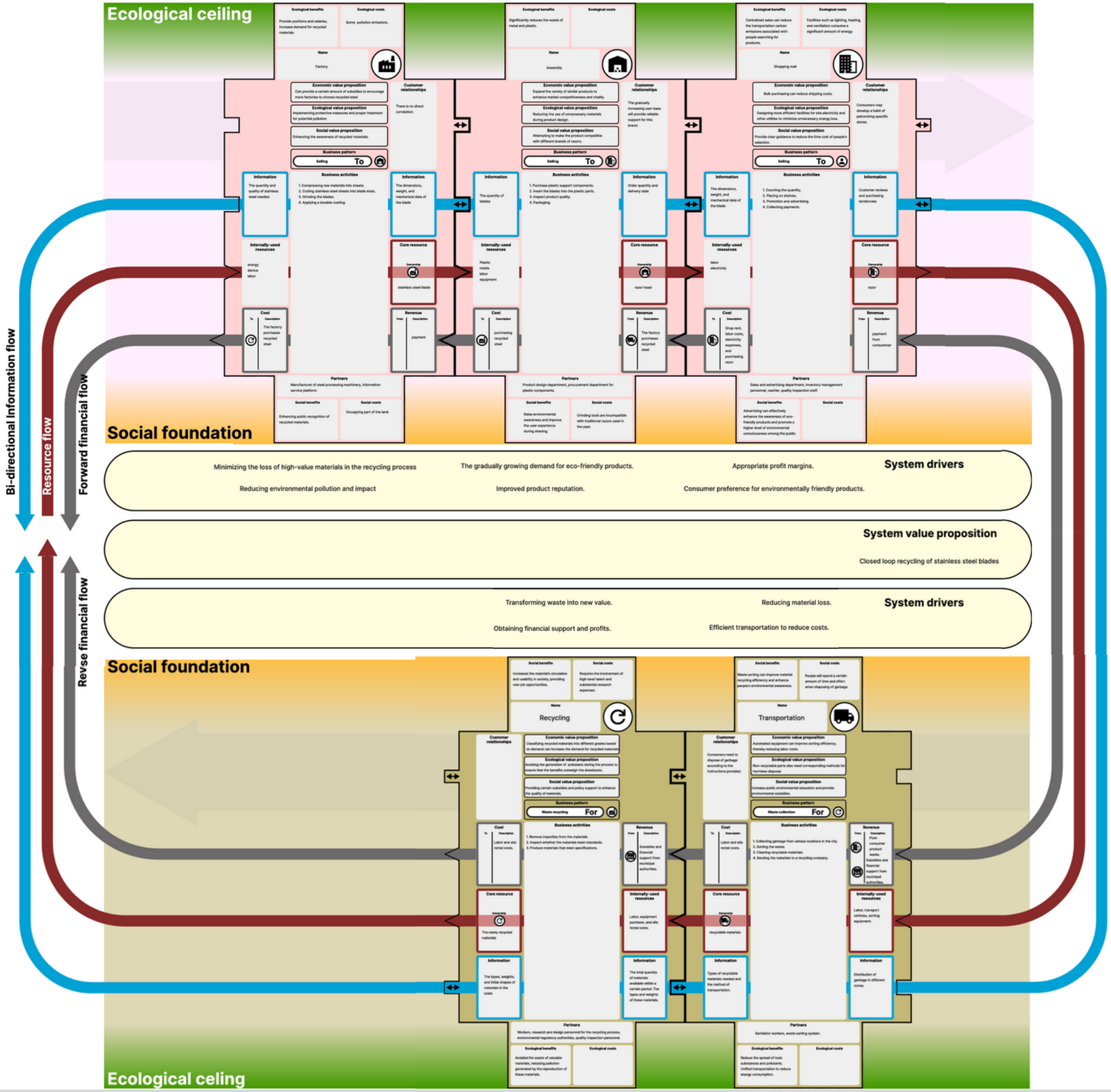







**References:** [39]: https://www.ecovative.com/ [40]: https://www.outokumpu.com/

HAOZHEN JIA




# Circular Ecosystem Business Model Canvas

Circular Ecosystem Business Model Canvas serves as a valuable tool for businesses seeking to embrace circular economy principles. It provides a structured approach to developing sustainable strategies, fostering innovation, and creating long-term value for both the business and the environment.

**SLOW** **NARROW** **CLOSE**

## Our Focus

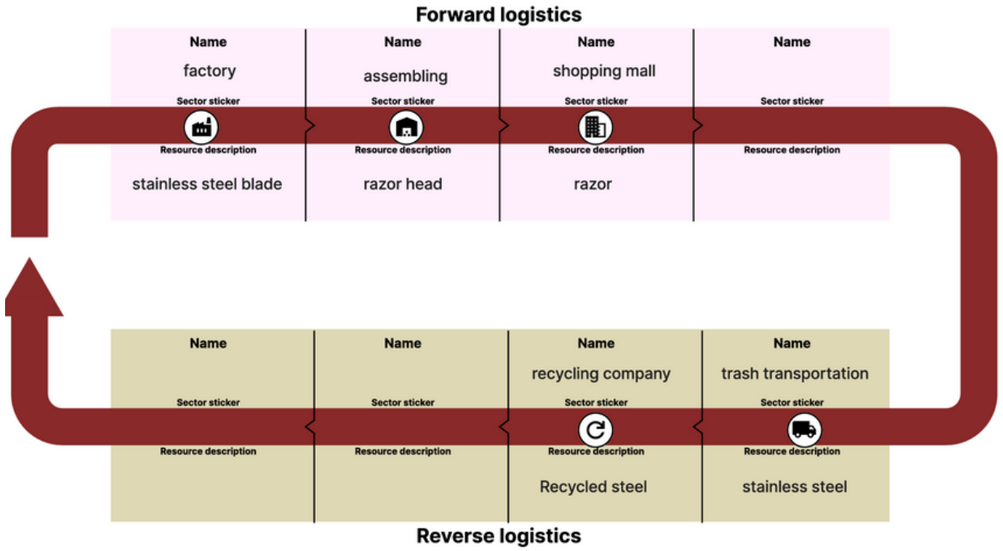
This is about the material cycle of stainless steel. Because it is the primary material for our products and has greater recycling value compared to packaging, we will address how to make the material undergo multiple production cycles.

In this section, we focus more on generating or increasing value, rather than analyzing issues directly related to user experience. Increasing value can allow the company to gain more potential profits and have a positive impact on society and the natural environment. This section provides the company with more possibilities and maximizes the development potential of this product.

## Identifying Risk and Opportunities

| | Revenue/benefit | cost | cost mitigation |
|---|---|---|---|
| **Finance** | • Enhancing public recognition of recycled materials.<br>• Raise environmental awareness and improve the user experience during shaving.<br>• Advertising can effectively enhance the awareness of eco-friendly products and promote a higher level of environmental consciousness among the public.<br>• Waste sorting can improve material recycling efficiency and enhance people's environmental awareness.<br>• Increased the material's circulation and usability in society, providing new job opportunities. | • The factory needs to pay the price for the recycled metal.<br>• The company needs to pay for the blades from the factory.<br>• The shopping mall needs to pay the company at a wholesale price.<br>• The eco-friendly transportation company needs to pay employee salaries.<br>• The recycling company needs to pay for patents and valuable equipment. | • Can provide a certain amount of subsidies to encourage more factories to choose recycled steel<br>• Expand the variety of similar products to enhance market competitiveness and vitality<br>• Bulk purchasing can reduce shipping costs.<br>• Automated equipment can improve sorting efficiency, thereby reducing labor costs.<br>• Classifying recycled materials into different grades based on demand can increase the demand for recycled materials. |
| **Ecology** | • Enhancing public recognition of recycled materials.<br>• Raise environmental awareness and improve the user experience during shaving.<br>• Advertising can effectively enhance the awareness of eco-friendly products and promote a higher level of environmental consciousness among the public.<br>• Waste sorting can improve material recycling efficiency and enhance people's environmental awareness.<br>• Increased the material's circulation and usability in society, providing new job opportunities. | • Some pollution emissions.<br>• Facilities such as lighting, heating, and ventilation consume a significant amount of energy. | • Implementing protective measures and proper treatment for potential pollution<br>• Designing more efficient facilities for site electricity and other utilities to minimize unnecessary energy loss.<br>• Reducing the use of unnecessary materials during product design.<br>• Non-recyclable parts also need corresponding methods for harmless disposal.<br>• Avoiding the generation of pollutants during the process to ensure that the benefits outweigh the drawbacks. |
| **Society** | • Enhancing public recognition of recycled materials.<br>• Raise environmental awareness and improve the user experience during shaving.<br>• Advertising can effectively enhance the awareness of eco-friendly products and promote a higher level of environmental consciousness among the public.<br>• Waste sorting can improve material recycling efficiency and enhance people's environmental awareness.<br>• Increased the material's circulation and usability in society, providing new job opportunities. | • Occupying part of the land<br>• Grinding tools are incompatible with traditional razors used in the past.<br>• People will spend a certain amount of time and effort when disposing of garbage.<br>• Requires the involvement of high-level talent and substantial research expenses. | • Enhancing the awareness of recycled materials.<br>• Provide clear guidance to reduce the time cost of people's selection.<br>• Attempting to make the product compatible with different brands of razors.<br>• Increase public environmental education and provide environmental subsidies.<br>• Providing certain subsidies and policy support to enhance the quality of materials. |

## System resource flow diagram




## Analys canvas

### Overall Strategy

- Emphasize public awareness and advertising for eco-friendly products.
- Implement efficient waste sorting and material recycling processes.
- Encourage the use of recycled materials through subsidies and market-driven strategies.

### Challenges

- Addressing pollution emissions and energy consumption in facilities.
- Managing costs related to land occupation, tool compatibility, and research expenses.

### Opportunities

- Creating new job opportunities through increased material circulation.
- Leveraging automation for efficiency and cost reduction.
- Aligning product design with environmental goals for societal and ecological benefits.

FRANCESCO RASTELLI

# Conditions:

## Societal Trends:

Changes in social trends can greatly affect the way that our product will be sold in the future. On one hand, we could have an increase in sales as people become increasingly aware of the impacts that their shopping trends have on the environment and so look for products that reflect those values accordingly. However, there might also be a shift and thus increase in facial hair for people so fewer of them would buy razors to shave themselves. Furthermore, the improvement in the quality and effectiveness of electric shavers has led to a societal shift which priorities these over the standard safety razor.





## Economic Conditions:

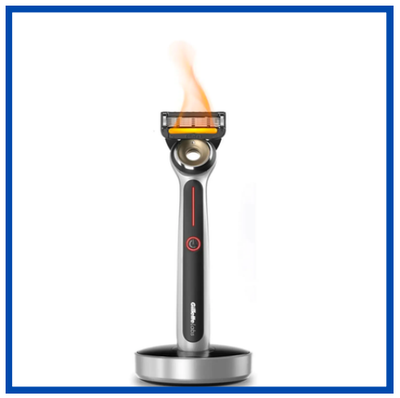
Economic factors such as inflation, interest rates and the overall state of the market directly impact how many razors are sold. As the market goes a dip and inflation rises, fewer and fewer people are inclined to buy things like disposable razors as they prioritise things which hold higher value to them, such as rent, bills and food. However, when the market is doing well and people have more disposable income, they become more likely to purchase things such as beauty products and often spend more money on them to purchase the higher-end and higher-quality ones.

## Weather:

Weather can affect the various aspects of the business, but most importantly the manufacturing and supply chain. Extreme weather events such as monsoons, flooding, earthquakes and landslides can affect the movement of goods, services and materials in their own respective ways. This can be even more seriously affected over long distances and the more complex a supply chain is the more likely it is to fee the impact of an event at the start of it.

## Technological Conditions:

Improvement in technologies from competitors would lead to consumers switching to their products. Current technologies for which sales are rising include electric shavers, which offer a lot of benefits compared to standard safety razors. However, we also have emerging and new technologies such as Gillette Labs Heated Razor which has “a state-of-the-art warming bar designed to deliver heat and comfort with every razor stroke” [a]. Although this technology is not widespread yet, the increase in comfort will definitely lead to it becoming a very popular product.



# Agreement:

## ICEA (The Ethical and Environmental Certification Institute) [b]

“ICEA is a stakeholder-owned certification organisation, with over 13,000 certified firms with strong ethical, environmental and social values” [c] and is one of the top certification bodies in Italy and Europe. Their certificate is used for many products ranging from sustainable fisheries, to cosmetic products, ethical trade and social accountability. For us to get this product, we would have to make sure that our product fulfils proper biological testing practices and has agreements with all our workers which ensures fair practices and takes them into account when making corporate decisions which may impact them.

## Nordic Swan EcoLabel [d]

The Nordic Swan EcoLabel is one of the hardest qualifications to acquire for a company. Their label offers: “a recipe on how to reduce the environmental impact from production and consumption of goods” and “credible, third party certified guidance for their consumers and professional buyers to choose goods and services that are among the environmentally best” [d]. Therefore, to get this qualification, you need to meet its requirements throughout the product’s entire lifecycle; from the raw materials, to manufacture, distribution, use of, recycling and disposal; but also the quality and durability of the product. To meet these, our product has used high-quality materials such as AISI 304 Stainless Steel which has good corrosion resistance, high tensile strength, and excellent edge retention; whilst using up to 50% recycled material. We have also looked at the supply chain and where materials are acquired and processed, and different companies we could look into that would ensure a proper end of life for our razor blades and product, such as SIMS Metal which would recycle the stainless steel blades.

## Agreements with companies in System Solutions Design:

Various agreements would have to be made with the different companies within the supply chain:

- To acquire raw materials to be processed, these being the steel, the whetstone and the mycelium packaging, agreements about the costs, distribution to the manufacturing and assembly plants and fair worker rights would have to be made.
- The manufacturing of the components under Kai Group and JC METALWORKS would also need similar agreements for the raw materials.
- The distribution process requires us to undergo contracts with companies such as Boots and Amazon regarding the price that they pay for our products and how they get distributed to them.
- Lastly, we would put in place agreements with the waste management companies so that we can ensure that our products are disposed of correctly.



**References:** [a] https://www.gillette.co.uk/gillettelabs/heated-razor-starter-kit/13486603.html. [b] https://icea.bio/en/who-we-are/ [c] https://directory.ifoam.bio/affiliates/500-icea-istituto-per-la-certificazione-etica-ed-ambientale [d] https://www.nordic-swan-ecolabel.org/official-nordic-ecolabel/

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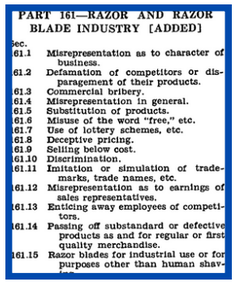
# Legislation:

## Lubricating strips on razors may contain suspected hormone disrupting chemicals: [a]

A new study from the Danish Consumer Council THINK Chemicals [b] has found that **4 in 11 razor brands** contain a chemical called **BHT** (butylated hydroxytoluene) which acts as an antioxidant and is a suspected **endocrine disruptor** (The endocrine system is made up of glands that make hormones and thus influences every cell, organ and functions of the body).

The evaluations for this study were based on a large number of lists which include "**Endocrine Disruptors"** [c] from the **European Commission**, Substances mentioned by the **National Allergy Research Centre** [d] and "**OPINION on Fragrance allergens in cosmetic products"** [e] by the **SCCS (Scientific Committee on Consumer Safety).** These are all highly trusted sources which have all mentioned BHT and have come to the conclusion that is can impact the endocrine system.

## Bribery of American legislators from razor companies: [e]



PART 161—RAZOR AND RAZOR BLADE INDUSTRY [ADDED]

Sec.
161.1 Misrepresentation as to character of business.
161.2 Defamation of competitors or disparagement of their products.
161.3 Commercial bribery.
161.4 Misrepresentation in general.
161.5 Substitution of products.
161.6 Misuse of the word "free," etc.
161.7 Use of lottery schemes, etc.
161.8 Deceptive pricing.
161.9 Selling below cost.
161.10 Discrimination.
161.11 Imitation or simulation of trademarks, trade names, etc.
161.12 Misrepresentation as to earnings of sales representatives.
161.13 Enticing away employees of competitors.
161.14 Passing off substandard or defective products as and for regular or first quality merchandise.
161.15 Razor blades for industrial use or for purposes other than human shav-

Whilst our company is not planning to bribe or defame any American legislators or other companies, we feel that it is important to mention such unfair trade practices as they may be relevant in the future. The first 13 sections ranging from **161.1** to **161.13** are about the different ways that bribery, misuse of terminology and deceptive practices may be used to influence a consumer into buying the product. However, the one that could be most relevant to us is **"161.14 Passing off substandard or defective products as and for regular or first quality merchandise"** [e] as it means that we need to ensure proper quality checks.

## Containment and disposal of razor blades according to OSHA: [f]

OSHA has put forward the **"1910.1030 Bloodborne pathogens"** [g] which "applies to all occupational exposure to blood or other potentially infectious materials" including razor blades and details how they should be disposed of accordingly. Section **1910.1030(d)(2)(vii)** specifies that **"Contaminated needles and other contaminated sharps shall not be bent, recapped, or removed"** [g] whilst section **1910.1030(d)(2)(viii)** states that **"contaminated reusable sharps shall be placed in appropriate containers until properly reprocessed".** These two clauses mean that for our product, the blades should be easy to remove after use in case of commercial use but also that we need to keep in consideration the fact that the razor cannot be used between different people.

# Standards:

## IS 13973 (1994): Shaving systems - Disposable razors [h]

Whilst "**this standard covers the requirements of disposable razors which is a self-sufficient unit with a blade fixed on the handle which may be twin type or single type used for shaving purpose"** [h], a lot of the dimensions are other requirements are relevant to the design of our product.

Starting with the material section, **5.3** states that **"The material of the blades shall be cold rolled stainless steel strip of medium thickness ( 0.1 ±0.005 mm ) as per IS 9294:1979."** This means that the thickness of our razor blades will be designed under the same standards.

For the workmanship and manufacture, **8.1** states that **"All the parameters of double edge such as cutting edge, straightness and parallelism, flexibility of blade, final bevel angle; of cutting edge, hardness and anti-corrosiveness shall conform to IS 7371:1982";** this will thus will apply on how our product is manufactured and the state in which it will arrive to the user.

Lastly, we can take inspiration from the performance test found under section **10** which gives us a guideline on how performance testing on the quality of our shaving can be looked at and used in order to improve the design and ensure it is good enough for the target market.

## ISO 10993: Standards for the biologic evaluation of medical devices [i]

ISO 10993 is made up of 23 different sections which detail different aspects regarding evaluation, testing and guidance on materials and components found within medical devices. Razors fall within this due to their use on the human body and thus are influenced by 3 key sections:

- **ISO 10993-10:2021** tests for **"skin sensitization"** [j] which is important to our product as it will directly come into contact with the skin. To do this, we want the materials within our products not to be irritants. A possible issue that might arise that will require future testing is that when sharpening, microscopic metal shards will form and sit on the blades which can definitely irritate the skin if not washed away properly before use.
- Leading on from this **"ISO 10993-22:2017 – Guidance on nanomaterials"** and **"ISO 10993-23:2021 – Tests for irritation"** [j] both offer guidance and testing methods so we can evaluate whether our product would irritate a user or not.

## FDA's Guidance on the use of International Standard ISO 10993-1 [k]

The FDA has put together a document which helps us understand the ISO above better. Under section **IV B. FDA USE OF ISO 10993-1**, they give us their 7 principles. However, the first 2 of these are the most important for us as they are the only to directly influence the design:

1. "The **selection of material(s)** to be used in device manufacture and its biocompatibility evaluation should initially take into account the likelihood of direct or indirect **tissue contact** and any available information for the materials of manufacture, for example, chemical formulation for each component material, including adhesives, **known and suspected impurities**, and constituents associated with **processing**." [k]
2. "The material(s) of manufacture, the device in its **final finished form**, and **possible leachable chemicals or degradation products** should be considered for their relevance to the overall biocompatibility evaluation of the device" [k]



**References:** [a] https://chemtrust.org/news/razors-suspected-edc/ [b]https://taenk.dk/kemi/english/avoid-razors-bht [c] https://single-market-economy.ec.europa.eu/sectors/cosmetics/cosmetic-products-specific-topics/endocrine-disruptors_en [d] https://www.videncenterforallergi.dk/ [e] https://ec.europa.eu/health/scientific_committees/consumer_safety/docs/sccs_o_102.pdf [e] https://tile.loc.gov/storage-services/service/ll/cfr/cf/r1/94/60/03/-T/16/CI/PI/61/cfr1946003-T16CIP161/cfr1946003-T16CIP161.pdf [f] https://www.osha.gov/laws-regs/standardinterpretations/2005-03-28 [g] https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1030 [h] https://law.resource.org/pub/in/bis/S01/is.13973.1994.pdf [i]https://www.rimsys.io/blog/iso-10993-biologic-evaluation-of-medical-devices [j] https://www.iso.org/ics/11.100.20/x/ [k] https://www.fda.gov/media/142959/download

# CONCLUSION: SOLUTION INNOVATION & FEASIBILITY

## Novelty & Unique Selling Points

### 1st Monomaterial Razor in the Market



Compared to top product in the market Gillette Fusion5:

**-65% Total Embodied Energy**

**-65.4% in Materials**

**-23% in Transport**

**-91% in EoL Potential**

The razor, made out of **304 Stainless Steel**, is a **corrosion resistant, high strength** and lasts for approx. **10 years**.

It is the only razor in the maket made out of the **same material for both blades and handle**, which hugely **facilitates recycling**.

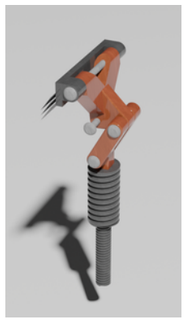
### 1st Razor Company Using Mycelium Packaging



As a **natural material from a renewable source**, it is easy to grow, needing very little water, light or space to produce at an industrial scale. It is also **high-performing, cost-competitive, and insulating protective** packaging solution.

This would be the **first implementation in the shaving industry**. Other companies such as **IKEA** have already implemented this packaging in furniture, proving the advantages of this solution.

### 1st Pushing Mechanism for Blade Sharpening



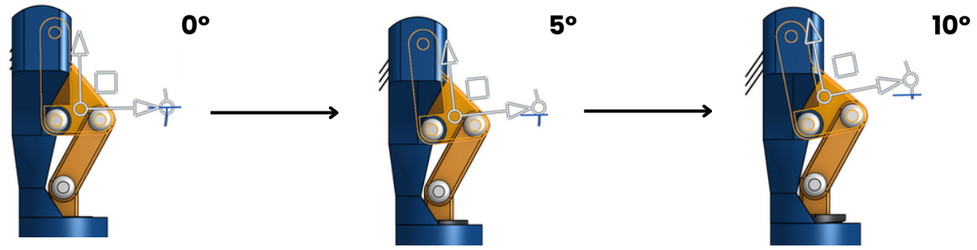


Razors have been released to the market with whestones to sharp the blades. However, **competitors have failed to address the decrease in lenght** and change in angle after sharpening.

The twisting L-Lever **increases the lifespan** of each blade, and **decreases waste** and the need for the user to purchase blades frequently. This is done while maintaining the ideal 35° shaving angle.

## Feasibility & Practicality

### Functional & Efficiently Manufactured




After running **CAD simulations in SolidWorks**, we proved that the mechanism is functional. The **twisting L-Lever is standard** and consists of screws, bars and bolts, thus it can be **easily and quickly produced** by our manufacturer Outokumpu.

### Viable & Sustainable Sourcing

304 Stainless Steel

outokumpu

Outokumpu is the only industry leader in the UK that has been **accredited** for sustainable sourcing and mining, and invited to the United Nations Climate Change Conference (**COP28**). Its reputation is attributed to its combination of product quality, diverse offerings, competitive pricing

Mycelium Packaging

ecovative

Ecovative is the **leader in the niche mycelium market**, backed up with **open source patents & strong partnerships** such as BESTSELLER. Ecovative crafts **biodegradable and compostable** packaging.

### Business Cost, Revenue & CSR

A sustainability-oriented company has higher production costs, and therefore has to ensure that profits are achieved to make it scalable in the long run.

Monetization Streams:

- **Initial high product cost** in exchange for longevity.
- **Catalogue** of blades, aftershave mosturizer, etc.
- **Ease to build consumer trust** due to long-lasting, eco-friendly product will lead to **word of mouth**.
- Sustainable **CSR values will boost partnerships** and facilitate loans or grants from investors

**- 93.3% Materials Cost**
(£15/kg PLA vs. £1/kg Stainless Steel)

**- 700 km Transportation per delivery**
(1000 km Berlin Gillette vs. 300km Sheffield MycoEdge)